

JERISON TIAN

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Education

University of Alberta

B.Sc. Mechatronics and Robotics Engineering, Co op

Expected 2030

Edmonton, AB

- Cumulative GPA 3.6/4.0. Engineering Entrance Scholarship, \$5,000. NSERC Undergraduate Student Research Award, \$9,100.

Technical Skills

- **Programming and Robotics:** C++, Python, ROS 2 Jazzy, CMake, Git, Linux Ubuntu, real time control, rosbag, QoS.
- **AI and Computer Vision:** PyTorch, Ultralytics YOLO, OpenCV, ONNX Runtime, CUDA, PnP pose estimation, keypoint detection, camera calibration.
- **Controls and Embedded:** Model Predictive Control, CasADi, IPOPT, system identification, state estimation, STM32, I2C, UART, MPU6050 IMU.
- **CAD and Hardware:** SolidWorks, 3D printing, Intel RealSense, Parrot Anafi and Bebop, Vicon, Arduino, SG90 servos.

Research Experience

Undergraduate Research Assistant, Autonomous Drone to Drone Tracking

May 2026 to Present

University of Alberta, Dr. Martin Barczyk, NSERC USRA

Edmonton, AB

- Developed and flight tested a ROS 2 autonomous drone tracking system integrating YOLO keypoint detection, PnP based 3D relative pose, Anafi telemetry, and independent X, Y, and Z control for real world target following.
- Rebuilt a monolithic Python tracking process into dedicated camera, detection, position estimation, controller, and command nodes, increasing the image and detection path from approximately 9 FPS to a consistent 30 FPS.
- Ported performance critical perception components to C++ and ONNX Runtime with NVIDIA CUDA, increasing detector throughput from approximately 15 FPS to 30 FPS.
- Designed three axis receding horizon MPC using CasADi and IPOPT with target motion estimation, self velocity compensation, predictive braking, asynchronous solving, and safety supervision.
- Experimentally identified axis specific Anafi dynamics from flight data, including approximately 0.32 s input delay and 2.48 s lateral time constant, then used the models to tune braking, command limits, arrival, and hold behavior.
- Extended the perception stack to a wide angle FPV camera using Vicon and VRPN ground truth at approximately 100 Hz and historical pose matching to compensate approximately 110 ms of video latency.
- Built an automated 8 keypoint YOLO pose labeling and QA pipeline and generated a session separated 428 image dataset for camera specific fine tuning with PyTorch and Ultralytics.
- Reduced effective state age to approximately 29 ms and achieved approximately 93.6% target freshness with no foreign target association in final phase testing through timing, observer, and target association improvements.

Projects

Predictive MPC Face Following Pan Tilt Camera

Completed

Independent Project

- Collected and annotated a custom image dataset and trained a PyTorch and YOLO single person detector to recognize one enrolled face, then exported the model to ONNX for real time C++ inference with OpenCV DNN.
- Designed and 3D printed a two axis pan tilt mechanism, integrated dual SG90 servos with Arduino serial control, and built independent pan and tilt Model Predictive Controllers running at approximately 20 Hz.
- Modeled each axis with separate response time, command gain, estimated delay, camera pixels per degree, velocity limits, and acceleration limits, with command queue delay compensation for future state prediction.
- Implemented target association, exponential smoothing, target motion estimation, predictive tracking, hysteresis, stationary hold, mechanical limits, target loss recovery, and controlled reset behavior.

Stabilized Steady Spoon Assistive Device

In Progress

Independent Project

- Designed a two axis active stabilization prototype using inertial sensing and servo actuation to counter involuntary hand tremor, combining mechanical design, embedded firmware, sensing, and iterative prototyping.
- Flashed and validated STM32 firmware and GPIO operation and integrated SG90 servo actuation with an external 5 V supply and shared ground for physical testing.
- Integrated an MPU6050 IMU over I2C and developed the hardware and firmware foundation for sensor fusion and closed loop actuator stabilization.

Leadership and Activities

- Badminton coach and competitive player. Royal Conservatory of Music Level 10 Piano.